The Market for Extortion and Protection

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Abstract

States worldwide face significant challenges in maintaining the Weberian concept of monopolizing the legitimate use of violence and ensuring the protection of individuals within their territories. The presence of non-state groups engaging in protection and extortion thwarts efforts to provide security, enforce property rights, address underdevelopment, and foster economic growth. This paper develops a model to analyze the dynamics of strategic competition in extortion and protection provision under the shadow of the state, examining the impact of various state policies on the outcomes of this violent competition.

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Project description

Latin American states face significant challenges in maintaining the Weberian concept of monopolizing the legitimate use of violence and protecting individuals within their territories. In Port-au-Prince, major gang coalitions have taken control of critical parts of the country, running kidnapping enterprises and charging informal fees for people and cargo being transported[1]. In Santiago, gangs engage in small-scale warfare over the control of downtown areas to extort and racketeer businesses[2]. In Mexico, the count of self-defense groups has reached a significant number, indicating a lack of state control[3]. Medellin has widespread evidence of "combos" providing order in exchange for the "vacuna."[4] In Rio de Janeiro, paramilitaries and various drug-trafficking organizations compete to control the favelas[5]. In El Salvador, the government has taken draconian measures to combat MS-13 and Barrio 18, which specialize in extorting the local population[6]. These examples illustrate the severe difficulties Latin American states face in asserting their authority and ensuring the safety of their inhabitants.

The prevalence of non-state groups specializing in illicit activities such as protection, racketeering, and extortion poses a core policy challenge when providing security to individuals and businesses, enforcing property rights, addressing underdevelopment and fostering growth. Current economic development theories underscore the necessity of a state that can effectively secure property rights and ensure the safety of its citizens and businesses. This project aims to enhance our theoretical understanding of the mechanisms when multiple non-state groups violently compete to extort the population and how this affects extortion's intensity under varying state policies. By examining the interactions between these groups and state authorities, we seek to shed light on how these dynamics impede economic development and explore potential policy solutions to mitigate their impact.

Most scholarly attention on these topics has concentrated on when criminal groups vie for control over other illicit markets such as drugs, oil, and gambling (Lessing 2015, 2017; Castillo and Kronick 2020; Castillo-Quintana 2024). However, the competition for territorial control to extort the population operates under different dynamics. This project has two primary objectives: first, to develop a game-

[1]https://www.nytimes.com/2024/03/07/world/americas/haiti-gangs-explainer.html

[2]https://www.mega.cl/programas/mucho-gusto/mega-investiga/169132-bandas-extranjeras-disputa-territorial-santiago.html

[3]https://www.diariolasamericas.com/america-latina/mexico-grupos-autodefensa-vencer-o-ceder-la-violencia-n5346407

[4]https://osf.io/preprints/socarxiv/3bncz

[5]https://brazilian.report/society/2024/04/17/drug-traffickers-militias-rio-territory/

[6]https://www.dw.com/es/el-salvador-suma-78000-presos-en-guerra-contra-pandillas/a-68676454

theoretical model that elucidates the dynamics of competition in the market of extortion and protection under the shadow of the state, and second, to complement the existing empirical literature on this subject. We aim to provide a deeper understanding of the mechanisms driving extortion and extortion-related violence and inform more effective policy responses.

Let's draw and compare two existing empirical papers that study this phenomenon to motivate some of the challenges underlying this research. Brown et al. (2024) study the extortion fees charged by Salvadorian gangs before and after a truce, they find that in times of peace, they charge higher fees to the truck drivers that dispatch merchandise to their territories. In contrast, Magaloni et al. (2020) find that territorial competition among Mexican criminal groups increases extortion. These apparently contradictory findings suggest the presence of distinct mechanisms influencing extortion, which may exert opposing effects. Why would competition increase extortion in Mexico while reducing it in El Salvador?

Contribution to the literature

While there is limited theoretical work on how competition for the market of protection influences extortion efforts over the population and a lack of theory on how various state policies might affect this relationship, this paper aims to fill this gap by drawing insights and parallels from existing literature on state formation, the economics of crime, the sociology of crime and criminal politics.

Frameworks from state formation serve as a crucial first building block for our work. Since Tilly (2017), the seminal notion that state-making resembles organized crime allows us to draw parallels between the violent competition among criminal groups and among states over the control of extortion of a given circumscribed population (Carneiro 1970). States competed for territory, leading to divisions where each surviving state emerged as the monopoly provider of protection and taxation (Lane 1958; Levi 1988). Following the typologies developed by Olson (1993), the outcome of a conflict between criminal groups determines which group will adopt the behavior of a roving bandit and which will establish itself as a stationary bandit.

Another strand of literature on which this paper will build is the economics of crime. Constructing on the frameworks of Schelling (1958) and Becker (1968), we will assume that when criminal groups focus on extorting civilian populations, they engage in rational cost-benefit analysis and strategically respond not only to state policies but also to civil responses and the actions of other criminal groups. When non-state criminal groups extort businesses and communities, it creates inefficiency because extortion costs can be passed along to customers and victims, particularly if the demand in the victim's market is

inelastic (Schelling 1984).

The project will build upon an important piece of literature where specialists in violence compete to get control of who extorts individuals or businesses. Skaperdas and Syropoulos (1993, 1995); Moselle and Polak (2001); Skaperdas (2001); Mehlum et al. (2002); Hafer (2006); Garfinkel and Skaperdas (2007); Brown et al. (2024) develop models where specialists of violence directly or indirectly compete over the right to extort economic activity. This class of models' main findings is that gang competition efficiency follows a U-shaped relationship with asymmetry in endowments, and the availability of more destructive weapons increases conflict inefficiency. Additionally, a higher return to banditry leads to an increase in the number of bandits and a decrease in welfare for both peasants and bandits, with monopolized crime leading to varying societal outcomes compared to disorganized crime—Moselle and Polak (2001) finds it results in better welfare outcomes. In contrast, Brown et al. (2024) finds the opposite. Furthermore, gang competition heightens violence, whereas collusion among gangs leads to higher levels of extortion.

McGuire and Olson (1996); Konrad and Skaperdas (1997) provide another class of models that focuses on the interaction between victims and the perpetrators of extortion. Their main findings are that monopoly bandit-state can enhance economic productivity by balancing taxation with public goods provision, highlighting the trade-off between immediate revenue extraction and long-term economic outcomes and that a gang with monopoly extortion engages in no extortion when victims are few, but as the number of victims increases, it becomes profitable to engage in extortion, and there will be expected losses from violence in larger populations.

Finally, Blattman et al. (2022) provide the only existing game theoretical framework where gangs respond to state policies. In their model, the state and a gang offer protection services without directly conflicting with each other, functioning as strategic substitutes. They also conducted a field experiment in Medellin, enhancing the security provision of civilian state presence. This led to varying outcomes depending on the initial level of state legitimacy.

A fundamental approach of this project is to integrate, in a tractable manner, the diverse mechanisms captured by existing models within the literature and to complement the existing empirical literature by capturing the scope conditions under which violent competition and extortion fees are positively and negatively associated. Additionally, it will provide a benchmark for the role of state policies in mediating this violence and extortion.

The model will be dynamic to allow the formation of peace agreements as an endogenous outcome. This integrated approach will enable a deeper analysis of the conditions under which violent actors adjust

their behaviors in response to state interventions, contributing to the literature to formulate policies to reduce violence and foster economic development.

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Appendix: a first draft of the model

Let's consider a discrete-time model where $t = 0, 1, ..., \infty$ denotes the period. In each period t, there are four different types of players: The government, firms, organized crime gangs, and citizens. They operate in the same geographical territory that has size $s \in \Re_+$. Players discount the future with the same discount rate $r > 0$.

Citizens Citizens are consumers of the goods sold by firms as well as the illegal goods sold by gangs, such as drugs, stolen and fake goods, etc. They also suffer the consequence of gang violence when this occurs. The mass of consumers is proportional to the size of the territory. The larger the territory, the larger the measure of consumers.

Firms Firms compete against each other in selling goods. Let's assume that at each time t there are N_t firms, indexed by $i = \{1, \ldots, n_t\}$, operating in the territory. Their profits at time t are given by $\pi(s, n_t)$, where $\pi_s > 0$ and $\pi_{n_t} < 0$.[7] Thus, the larger the territory and the less intense the competition, the larger the profits. At each period t, firms face an opportunity cost given by $\bar{\pi}$. To take this opportunity, the production process in the current territory must be closed down.

Gangs At each period t, there are $g \in \{0, \ldots, G_t\}$ organized crime gangs in the territory. Gangs make profits from their illegal business given by $v(s, G_t)$, with $v_s > 0$ and $v_{G_t} < 0$. They can also fight with other gangs to gain more control over the territory. They can fight one other gang in each period t. Fighting costs them f and they are successful with probability $p_g \in (0, 1]$. Fighting imposes a cost on citizens equal to F.

When gang g fights gang g', they expel gang g' from the territory only when gang g succeeds and g' fails, when both succeed or both fail, both gangs remain in place.

When a gang is expelled from the territory at time t, it ceases to exist from time $t + 1$ on. There is a fixed entry cost K to form a gang. For the sake of simplicity, we assume that when a gang forms, it does not take gang members from the existing gangs and, therefore, it does not affect its productive capacity. They recruited members from the continuum of citizens. Thus, citizens can be consumers and gang members at the same time.

Because the business profits from illegal business fall with the number of gangs and fighting other

[7] These are the result of an un-modeled market competition game where they chose a strategic variable such as prices or quantities. We assume that that market game has a continuous selection.

gangs is costly, there is a free-rider problem in the fighting between gangs. Every gang prefers that other gangs fight between them since the non-fighting gangs get a higher business return when one gang is expelled from the territory.

At each period t, gangs engage in extortion activities by charging firms an extortion fee e_g in exchange for protection and charging E to other firms that gang g is not protecting and that are either unprotected or the gang who is protecting them fail to do so. When a gang successfully protects a firm, no gang can extract an extortion fee.

When firm i pays the extortion fee e_g to gang g, this provides the firm with protection at a cost $c_g a_g^i$, where $a_g^i \in \Re_+$ is the protection effort. The protection is successful with probability $q(a_g^i, A_{-g}^i; \alpha)$, where A_{-g}^i is the competing gangs' effort profile to make gang g's protection strategy applied to firm i fail and α is a parameter that measures the protection effort made by the State. The protection probability $q(a_g^i, A_{-g}^i; \alpha)$ increases with a_g^i and falls with A_{-g}^i. When protection succeeds, other gangs cannot extort the firm. The marginal cost of the effort to fight gang g' is C_g. Thus, the total cost is $C_g A_g$, where $A_g = (\ldots,, A_g^{i-1}, A_g^{i+1}, \ldots)$.

A gang extracts E from each firm that is not successfully protected. Hence, the expected extortion fee that gang g gets $E(A_{-g}) = \sum_{i' \neq i}(1 - q(a_{g'}^{i'}, A_{-g'}^{i'}; \alpha))E$, where $A_{-g} \equiv (\ldots, A_{g-1}^i, A_{g+1}^i, \ldots)$ and $A_{-g}^i = (\ldots, A_{g-1}^i, A_{g+1}^i, \ldots)$.

Thus, a firm that is not successfully protected pays extortion fees an amount $(G-1)E$ when it pays gang g for protection and pays GE when it does not pay any gang for protection.

The protection technology is such that $q(a_g^i, 0; \alpha) = 1$ whenever either $\alpha > 0$ or $a_g^i > 0$ or both. That is, if no gang extort firm i, this is successfully protected either by the gang hired or the State or both.

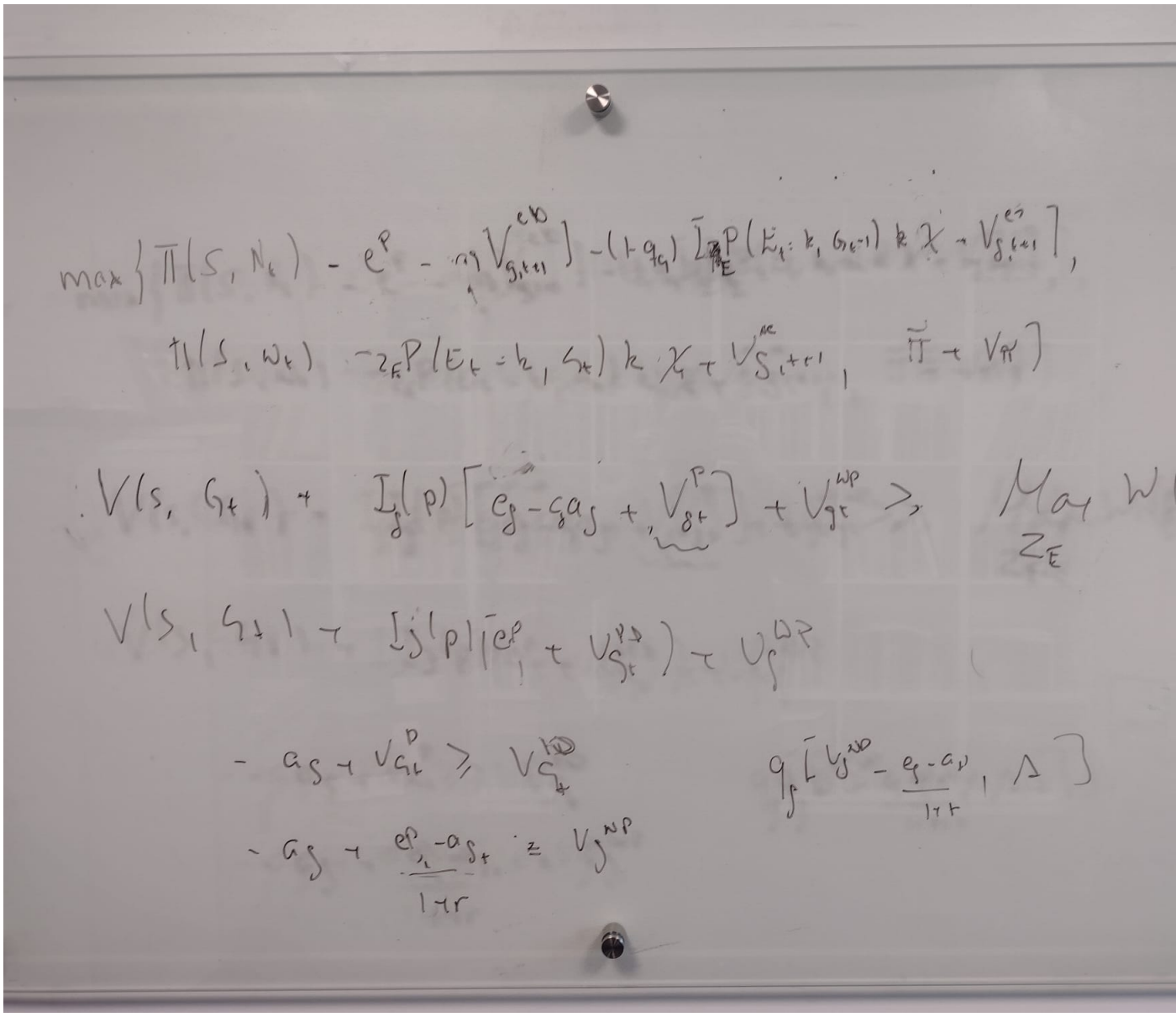
Gangs choose to fight and offer protection simultaneously and firms after that can either accept or reject any protection offer.

The State The main role of the estate is to protect firms and citizens. The state can thwart gangs by investing in policing the territory and in policy intelligence. The protection of the State makes gangs' extortion less likely to succeed. In particular, when firm i does not pay for protection or pays for it to gang g but this reneges on it and does not provide it, the probability of being protected is given by $q(0, A_{-g}^i; \alpha) \geq 0$. The state can affect $q(a_g^i, A_{-g}^i; \alpha)$ by choosing α at cost $K\alpha$, with $q(a_g^i, A_{-g}^i; \alpha)$ increasing in α. The costs are borne out by citizens.

Welfare in the absence of gangs is given by $W(s, N_t)$, with $W_s(s, N_t) > 0$ and $W_{N_t}(s, N_t) \gtreqless 0$. This is meant to capture that there could be socially excessive entry, as in the Cournot model, or lack of it as in

other oligopoly models. To the welfare, we have to subtract the cost of the violence when it takes place and the cost of policing and preventing extraction.

$$\max\left\{\Pi - e_g + q_g \cdot \left[V^{e_g}_{g,t+1}\right] - (1-q_g)\left[P(E_t = k, t-1)ke_g +) + V^{e_g}_{g,t+1}\right], \Pi - P(E_t = k, t-1)ke_g + V^{e_g}_{g,t+1}, \frac{\tilde{\Pi}}{1-\delta}\right\}$$



IC for the stationary gang. If gang g reneges on its protection promised to firm i, then neither firm i nor firm i' trusts gang g in the future. Thus, gang g cannot extract a protection fee from any firm forever thereafter. Thus, gang g can only extract E from firms whose protection failed. Firm i can pay gang g'

for protection from the next period on. Thus, firm i will be protected with probability $q(a_g^i, A_g^i)$.

$$\frac{e_g + E(A_{-g}) - c_g a_g^i - C_g A_g}{1-\delta} \geq e_g + (1 - q(0, A_{-g}^i; \alpha))E + E(A_{-g}) - C_g A_g + \frac{\delta}{1-\delta} V_g^P,$$

where the payoff during the punishment phase is

$$V_g^P \equiv \max_{A_g \in \Re_+^n} \{E(A_{-g}) + (1 - q(a_{g'}^i, A_{-g'}^i)))E - C_g(A_g + A_g^i)\}$$

This can be written as follow

$$\frac{\delta}{1-\delta} e_g \geq \frac{c_g a_g^i}{1-\delta} + (1 - q(0, A_{-g}^i; \alpha))E - \frac{\delta}{1-\delta}(E(A_{-g}) - C_g A_g) + \frac{\delta}{1-\delta} V_g^P$$

$$\iff$$

$$e_g \geq \frac{c_g a_g^i - \delta(E(A_{-g}) - C_g A_g) + (1-\delta)((1 - q(0, A_{-g}^i; \alpha))E + \delta V_g^P}{\delta}$$

IC for the firm: If firm i reneges on the protection fee promised to gang g, then gang g and no other gang g' trust firm i. Therefore, firm i cannot have access to gang protection forever thereafter. The only protection is the one coming from the State, which means that the firm is protected with probability $q(0, A_g^i; \alpha)$.

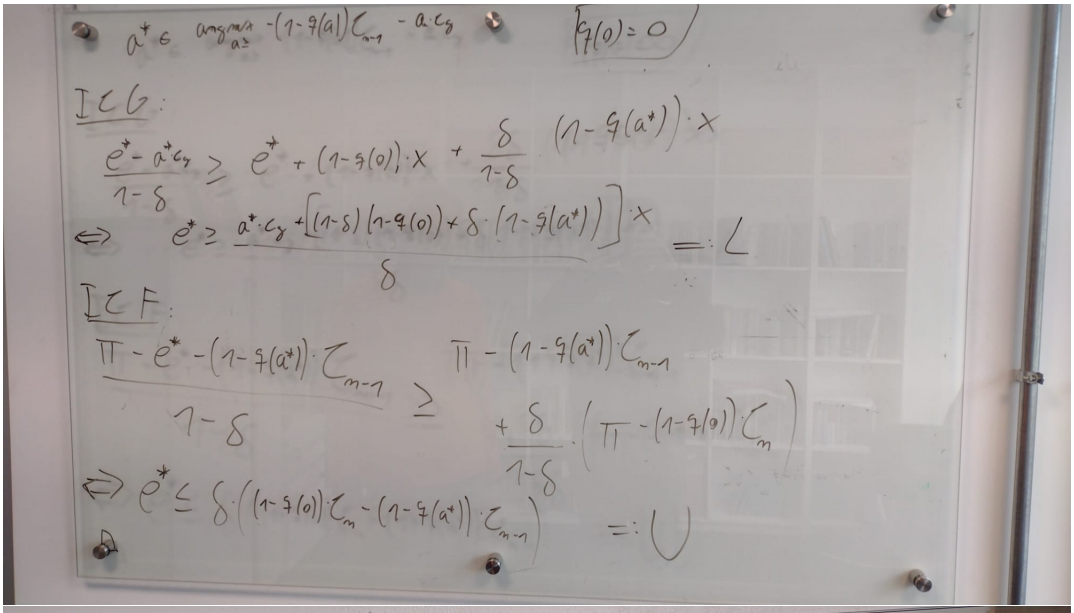
$$\frac{\Pi - e_g - (1 - q(a_g^i, A_{-g}^i; \alpha))(G-1)E}{1-\delta} \geq \Pi - (1 - q(a_g^i, A_{-g}^i; \alpha)(G-1)E + \delta \frac{\Pi - (1 - q(0, A_{-g}^i; \alpha))GE}{1-\delta}$$

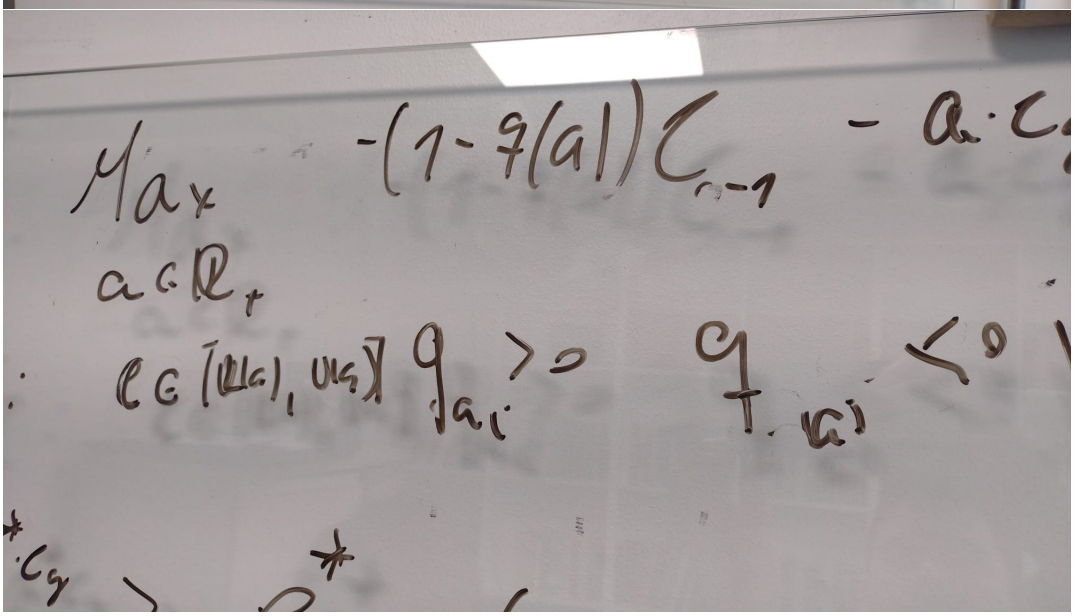
$$\iff$$

$$e_g \leq \delta\big((1 - q(0, A_{-g}^i; \alpha))G - (1 - q(a_g^i, A_{-g}^i; \alpha))(G-1)\big)E$$

Protection by gang g and the payment by firm i is incentive compatible if and only if the protection fee satisfies the following

$$e_g \in \Bigg[\frac{c_g a_g^i - \delta(E(A_{-g}) - C_g A_g) + (1-\delta)((1 - q(0, A_{-g}^i; \alpha))E + \delta V_g^P}{\delta},$$
$$\delta((1 - q(0, A_{-g}^i; \alpha))GE - (1 - q(a_g^i, A_{-g}^i; \alpha))(G-1)E)\Bigg]$$





The partnership between a gang and a firm maximizes the joint surplus

$$\max_{(a_g^i, A_g)\in\Re_+^n} \Pi - e_g - (1-q(a_g^i, A_{-g}^i;\alpha))(G-1)E + e_g + E(A_{-g}) - c_g a_g^i - C_g A_g$$

subject to

$$F(a_g^i, A_{-g}^i) \geq 0.$$

where

$$F(a_g^i, A_{-g}^i) \equiv \delta((1-q(0, A_{-g}^i;\alpha)GE - (1-q(a_g^i, A_{-g}^i;\alpha))(G-1)E) -$$
$$\delta^{-1}\big(c_g a_g^i - \delta(E(A_{-g}) - C_g A_g) + (1-\delta)((1-q(0, A_{-g}^i;\alpha))E + \delta V_g^P\big).$$

Let λ_g be the Lagrange multiplier for gang g's problem. Then the first-order condition for protection effort is given by

$$
\begin{aligned}
a_g^i : & q_{a^i}(a_g^i, A_{-g}^i; \alpha)(G-1)E - c_g + \lambda_g \delta^{-1} \left(\delta^2 q_{a^i}(a_g^i, A_{-g}^i; \alpha)(G-1)E - c_g \right) = 0, \\
& \lambda_g F(a_g^i, A_{-g}^i) = 0, \ \lambda_g \geq 0, \ and \ F(a_g^i, A_{-g}^i) \geq 0, \\
A_g^{i'} : & -q_{A_g^{i'}}(a_{g'}^{i'}, A_{-g'}^{i'}; \alpha))E - C_g + \lambda_g \left(-q_{A_g^{i'}}(a_{g'}^{i'}, A_{-g'}^{i'}; \alpha))E - C_g \right) = 0.
\end{aligned}
$$

We need to deal with the following challenges.

i) To prove the existence of SPE because there are g gangs and n firms playing the repeated game (must be done)

ii) To incorporate the State at the beginning of the game (at time 0). The state chooses α that affect the protection technology for all (i, g) pairs (proposed challenge)

iii) To determine the optimal e_g and E in the symmetric equilibrium by a free entry equilibrium condition. Gangs enter up to the point where it is no longer profitable to do so (proposed challenged).